Assignment 2 - Part 3

2AMS11 Survival Analysis for Data Science — Group Assignment, Part III — October 2026

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1 Practical context

Acute liver failure (ALF) is a sudden loss of liver function. Therapeutic plasma exchange (TPE) replaces the patient's plasma to remove toxins, prevent brain swelling and support the other organs. It buys time for the liver to regenerate by itself. **Event:** failure, i.e. death or liver transplant, whichever comes first. Surviving without a transplant ("transplant-free survival") is a success. **Time origin:** the first TPE cycle; time is measured in days. **Population:** adults with ALF who are treated with TPE. **Why survival analysis:** the outcome is a time to an event, and many patients are still alive without a transplant when follow-up ends, so their times are right-censored.

2 Survival model

We use a **log-logistic** model ($\log T$ follows a logistic distribution). When its shape parameter $\gamma > 1$, the hazard first rises and then falls. This matches ALF: the risk is highest in the first days, while the liver is failing (a "dangerous peak"). Patients who survive this period usually recover, so their risk drops.

3 Covariates

- D, **days from diagnosis to first TPE**: later treatment lets toxins build up and damages more organs ⇒ shorter survival.
- E, **etiology**: 1 = cause that also damages other organs (e.g. accident, heart failure), 0 = liver-only cause (e.g. drug, virus). Damage outside the liver ⇒ shorter survival.
- A, **age**: older bodies regenerate the liver more slowly ⇒ shorter survival.
- K, **number of comorbidities** (e.g. diabetes, immune disorders): these slow recovery ⇒ shorter survival.

Relationships: older patients have more comorbidities and are more likely to have a heart-related cause (poor blood flow to the liver). Age therefore affects E, K and T, which makes it a *confounder* for the effects of E and K (e.g. $K \leftarrow A \rightarrow T$). D is assumed to be independent of the others. All covariates are measured at the first TPE cycle.

4 Censoring

The data are **right-censored** for two reasons. (i) *Type I:* follow-up ends at $\tau = 90$ days, and patients who are alive without a transplant at that point are censored. (ii) *Loss to follow-up*, e.g. transfer to another hospital. We observe $Y = \min(T, C)$ and $\Delta = I\{T \le C\}$. We assume **uninformative censoring**: C is independent of T given the covariates. This holds for (i) by design. It may fail for (ii) if very sick patients are more often transferred; survival would then be overestimated. A transplant counts as a failure, not as censoring. Otherwise censoring would be informative, because the sickest patients are the ones who receive a transplant.

5 Data-generating mechanism

Model. Each patient's survival time T is log-logistic:

$$S(t) = \frac{1}{1 + (t/\lambda)^{\gamma}}, \qquad \lambda = \exp\left(\beta_0 + \beta_D D + \beta_E E + \beta_A A + \beta_K K\right).$$

$S(t)$ is the probability of being alive without a transplant after t days. λ is the patient's median survival time, since $S(\lambda) = \frac{1}{2}$. Each covariate changes λ: one unit more multiplies the median by the time ratio $\phi = e^{\beta}$. γ is the shape, and $\gamma > 1$ gives the early peak in the hazard. (Lecture form: $1/(1 + \tilde{\lambda} t^{\gamma})$ with $\tilde{\lambda} = \lambda^{-\gamma}$.)

Simulation steps for each patient: (1) draw D, A, K, E from Table 1; (2) compute λ; (3) draw $T = \lambda\,(U/(1-U))^{1/\gamma}$ with $U \sim U(0,1)$, which is the inverse-CDF method; (4) draw $C = \min(90, L)$ with $L \sim U(0, c_{\max})$; (5) record $Y = \min(T, C)$ and $\Delta = I\{T \le C\}$. We will use $n \in \{200, 500\}$ patients and $R = 2000$ simulation runs.

How we chose the numbers. The values are our own plausible assumptions; they are not estimated from data. The signs of the β's follow Section 3. Etiology has the largest effect because the cause of ALF is a strong predictor of outcome in registry data; the other effects are moderate. The covariate distributions are chosen to give realistic values, and age raises both the mean of K (0.6 at age 45, 1.6 at age 65) and $P(E = 1)$ (23% vs. 50%). $\beta_0 = 5.8$ and $\gamma = 1.1$ are chosen so that a reference patient ($D = 4$, $E = 0$, $A = 45$, $K = 0$) has a median of $e^{5.8 - 0.4 - 0.9} = e^{4.5} \approx 90$ days and a hazard peak at $\lambda\,(\gamma - 1)^{1/\gamma} \approx$ day 11.

Table 1: Covariates and true effects. $\phi = e^{\beta}$ is the time ratio per unit; $\phi < 1$ means shorter survival.

Covariate	Distribution	β	ϕ
D: days from diagnosis to TPE	Gamma(2, scale 2), mean 4	−0.10	0.90
A: age (years)	$\mathcal{N}(45, 15^2)$, capped to $[18, 85]$	−0.02	0.98
K: number of comorbidities	Poisson, mean $e^{-2.75 + 0.05A}$	−0.25	0.78
E: damage outside the liver	Bernoulli, $P(E{=}1) = \frac{1}{1+e^{3.9 - 0.06A}}$	−0.70	0.50

6 Research questions

RQ1. How accurately (bias, variance, MSE) does the log-logistic model estimate the true β's, and how does this change with sample size ($n = 200$ vs. 500) and loss to follow-up (5% vs. 10%)? **RQ2.** How biased are the estimates if we (a) fit a Weibull model, which cannot capture the early peak, or (b) leave out the confounder age?

7 Simulation algorithm

Our goal is to generate synthetic datasets from the Part I model and then fit parametric models to the observed values.

Setup. Fix the true parameters (see Table 1, Section 5) and $\gamma = 1.1$, $\tau = 90$ days, $n \in \{200, 500\}$, loss to follow-up $p \in \{5\%, 10\%\}$ and $R = 2000$ runs. For each p, find $c_{\max}$ numerically (`uniroot`, on 10^6 simulated patients) such that $P(L < \min(T, \tau)) = p$, with $L \sim U(0, c_{\max})$. This is the probability that a patient is lost to follow-up before their event or the 90-day cutoff. Then, for each of the four scenarios (n, p) and each run $r = 1, \ldots, R$: a) **Generate covariates** D, A, K, E for n patients from the distributions in Table 1. b) **Generate event times** $T = \lambda\,(U/(1-U))^{1/\gamma}$ with $U \sim U(0,1)$ and $\lambda = \exp(\beta_0 + \beta_D D + \beta_E E + \beta_A A + \beta_K K)$ (inverse-CDF method). c) **Generate censoring times** $C = \min(\tau, L)$ with $L \sim U(0, c_{\max})$. d) **Construct the observed dataset:** $Y = \min(T, C)$ (`time`) and $\Delta = I\{T \le C\}$ (`status`). e) **Fit the models** with `survreg`: M1 log-logistic with all covariates (RQ1), M2 Weibull with all covariates (RQ2a), M3 log-logistic without age (RQ2b). f) **Store** $\hat{\beta}$, the scale ($= 1/\hat{\gamma}$ for the log-logistic model) and the standard errors of each fit. Fits that fail or do not converge are skipped and counted. g) **Repeat** for all $R = 2000$ runs and four scenarios, then compute the performance measures. In the code, all $n \cdot R$ patients of a scenario are generated at once and split into runs by `run_id`.

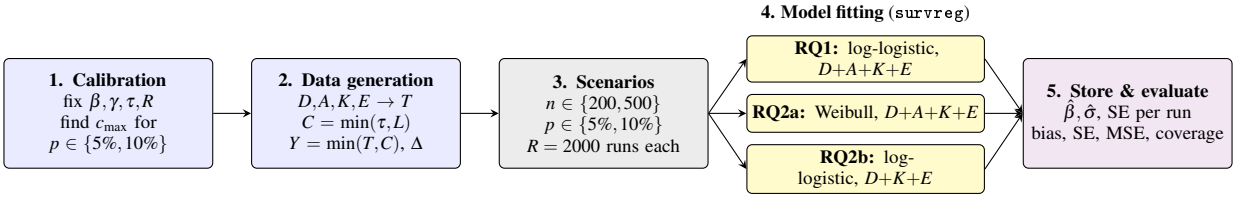


Figure 1: Simulation workflow: calibration of $c_{\max}$, data generation, the four scenarios, the three model fits and the evaluation.

8 Performance measures

We evaluate the estimates of $\beta_D, \beta_E, \beta_A, \beta_K$, with β_E as the main target. In each of the $R = 2000$ runs we store the estimate $\hat{\beta}$ and its standard error. With $\bar{\beta}$ the average estimate over all runs, we compute: **Bias** $= \bar{\beta} - \beta$: is the estimate systematically wrong? **Empirical SE** = the standard deviation of the estimates: how much does the estimate vary between runs? (Variance = SE^2.) **MSE** = the average of $(\hat{\beta} - \beta)^2 \approx \text{Bias}^2 + \text{Variance}$: the total error. **Coverage** = the share of 95% confidence intervals that contain the true β: are the standard errors reliable? It should be close to 95%. For each scenario we also report the observed censoring rate and the number of runs that failed to converge (these are excluded). Both the log-logistic and the Weibull fits are AFT models fitted with `survreg`, so their coefficients are on the same scale and can be compared with the same true values. When age is left out, only $\beta_D, \beta_E, \beta_K$ are evaluated.

9 Alternative (incorrect) analyses

(a) Wrong distribution: fitting a Weibull model. We fit a Weibull AFT model instead of the log-logistic one. *Why it is wrong:* the Weibull hazard can only increase, decrease or stay constant. It cannot rise and then fall, so it cannot capture the early "dangerous peak" in the true hazard. **(b) Omitting an important covariate: leaving out age.** We fit the correct log-logistic model, but without age. *Why it is wrong:* age is a confounder. It increases the number of comorbidities (K) and the chance of an extra-hepatic cause (E), and it also shortens survival directly.

10 Hypotheses

RQ1: correct log-logistic model. H1 (bias). The estimated effects will be very close to the true values in all four scenarios, and any small bias at $n = 200$ will be even smaller at $n = 500$. The fitted model is the model that generated the data and censoring does not depend on the survival time, so the estimates are correct on average. **H2 (sample size).** The empirical SE will drop by about one third and the MSE will fall to roughly 40% of its value at $n = 200$. More patients give more information, and with a bias close to zero the MSE is almost entirely variance. **H3 (loss to follow-up).** Going from 5% to 10% loss will make the estimates slightly less precise but not biased, an effect much smaller than that of sample size. Extra random censoring only removes some observed events without pushing the estimates in a direction, and many patients are already censored at day 90.

RQ2a: Weibull instead of log-logistic. H4. The Weibull model will clearly misestimate the intercept and the shape, while the covariate effects will be only slightly biased. The two models differ only in the assumed shape, so most of the misfit is absorbed by the intercept and shape. The Weibull hazard can only go up or down, so it cannot reproduce the early peak; because many patients are censored, the covariate effects will still be somewhat affected. **H5.** The Weibull bias will not go away with more patients while its variance will, so it will look relatively worse at $n = 500$ than at $n = 200$. A wrong model precisely estimates the wrong thing: more data makes the estimate more stable but not more accurate. **H6.** The Weibull model will be somewhat more biased at 10% loss than at 5%. With more censoring the estimates depend more on the assumed (wrong) shape of the distribution.

RQ2b: leaving out the confounder age. H7. Comorbidities and damage outside the liver will look more harmful than they really are. In our data older patients have more comorbidities, more often have damage outside the liver and survive shorter, so without age part of its harmful effect is wrongly attributed to K and E. **H8.** The effect of days to treatment will stay close to its true value but be estimated slightly less precisely. Days to treatment was generated independently of age, so age does not confound it; the survival variation caused by age becomes unexplained noise, which reduces precision but does not push the estimates in a direction.

11 Results

We ran the simulation from Part II (seed 123, 2000 runs per scenario). The observed censoring was 43% (5% loss to follow-up) and 45% (10% loss). Table 2 reports bias, empirical SE, MSE and coverage for β_E and β_K at 5% loss; the 10% loss results differ by less than 0.01 in SE. Figure 2 shows the distribution of the estimates in all four scenarios.

Correct analysis (M1). Bias is between -0.009 and $+0.004$, coverage is 94–95%, and the MSE of β_E is 0.081 at $n = 200$ and 0.030 at $n = 500$.

Incorrect analyses. *M2 (Weibull):* bias is $+0.08$ to $+0.10$ for β_E and $+0.03$ to $+0.04$ for β_K; coverage is 87–92%. *M3 (no age):* bias is -0.16 to -0.17 for β_E and -0.12 for β_K; coverage for β_K is 78% at $n = 200$ and 54% at $n = 500$. For β_D (not in the table), M3 has a bias below 0.001 and 94–95% coverage.

Table 2: Results at 5% loss to follow-up over 2000 runs. SE = empirical standard error; Cov. = share of 95% intervals that contain the true value.

		Etiology β_E (true -0.70)				Comorbidities β_K (true -0.25)			
Model	n	Bias	SE	MSE	Cov.	Bias	SE	MSE	Cov.
M1 (correct)	200	-0.009	0.284	0.081	95%	$+0.001$	0.126	0.016	95%
	500	$+0.004$	0.173	0.030	95%	$+0.001$	0.080	0.006	94%
M2 Weibull	200	$+0.078$	0.264	0.076	92%	$+0.031$	0.122	0.016	91%
	500	$+0.097$	0.163	0.036	89%	$+0.036$	0.078	0.007	87%
M3 no age	200	-0.170	0.276	0.105	90%	-0.124	0.110	0.027	78%
	500	-0.158	0.169	0.053	86%	-0.124	0.069	0.020	54%

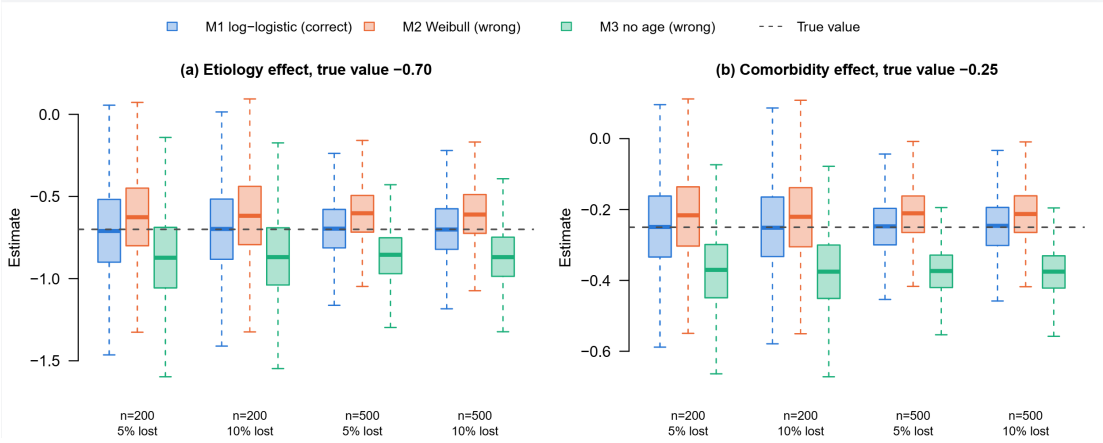


Figure 2: Estimates of β_E and β_K over 2000 runs (boxes: middle 50%; dashed line: true value). Only M1 is centred on the truth.

12 Evaluation

13 Comparison with expectations

14 Conclusions

15 Limitations

Use of AI tools

[TODO:]

Contribution statement

- Avram Andrei: **[TODO:]**
- Mihaiescu Cristian: **[TODO:]**
- Szelestey Adam: **[TODO:]**
- Vrincianu Darius: **[TODO:]**
- Zeng Qin Tao: Results (performance measures code, tables and figure).

All group members confirm that this contribution statement is accurate.

References

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16 Appendix

Software implementation

Condensed R code (left: data generation and censoring; right: model fitting and storage).

```r
set.seed(123)
beta <- c(b_0 = 5.8, b_d = -0.1, b_a = -0.02,
          b_k = -0.25, b_e = -0.7)
gamma_val <- 1.1; tau <- 90; sim_runs <- 2000
loss_vec <- c(0.05, 0.10)

generate_data <- function(m) {          # covariates
  d <- rgamma(m, shape = 2, scale = 2)
  a <- pmin(pmax(rnorm(m, 45, 15), 18), 85)
  k <- rpois(m, exp(-2.75 + 0.05 * a))
  e <- rbinom(m, 1, plogis(-3.9 + 0.06 * a))
  data.frame(days = d, age = a, nr_comor = k, damage = e)
}
draw_t <- function(x, beta, gamma_val) { # event times
  lambda <- exp(beta["b_0"] + beta["b_d"] * x$days +
    beta["b_a"] * x$age + beta["b_k"] * x$nr_comor +
    beta["b_e"] * x$damage)
  u <- runif(nrow(x))
  lambda * (u / (1 - u))^(1 / gamma_val)
}
# calibrate c_max: P(L < min(T, tau)) = target loss
find_cmax <- function(target, m, cutoff) {
  times <- draw_t(generate_data(m), beta, gamma_val)
  min_t <- pmin(times, cutoff)
  f <- function(cm) mean(pmin(min_t, cm) / cm) - target
  uniroot(f, c(5, max(times)), tol = 0.5)$root
}
cmax <- lapply(loss_vec, find_cmax, m = 1e6, cutoff = tau)

generate_cohort <- function(n, runs, cmax) {
  m <- n * runs                         # all runs at once
  x <- generate_data(m)
  t_event <- draw_t(x, beta, gamma_val)
  cens <- pmin(tau, runif(m, 0, cmax))  # C = min(tau, L)
  stat <- as.integer(t_event <= cens)
  rid  <- rep(seq_len(runs), each = n)
  c_n  <- tapply(1L - stat, rid, sum)   # per-run censor count
  data.frame(run_id = rid, x,
             time = pmin(t_event, cens), status = stat,
             censored = as.integer(c_n[rid]))
}
simulated_data <- list(
  n200_loss05 = generate_cohort(200, sim_runs, cmax[[1]]),
  n200_loss10 = generate_cohort(200, sim_runs, cmax[[2]]),
  n500_loss05 = generate_cohort(500, sim_runs, cmax[[1]]),
  n500_loss10 = generate_cohort(500, sim_runs, cmax[[2]]))
saveRDS(simulated_data, "simulated_data.rds")
```

```r
library(survival)
simulated_data <- readRDS("simulated_data.rds")

# Per-run empirical loss to follow-up and total censoring
loss_run <- lapply(simulated_data, function(d)
  tapply(d$status == 0 & d$time < 90, d$run_id, mean))
cens_run <- lapply(simulated_data, function(d)
  tapply(d$status == 0, d$run_id, mean))
censoring_summary <- data.frame(
  loss_mean = vapply(loss_run, mean, numeric(1)),
  loss_sd   = vapply(loss_run, sd, numeric(1)),
  cens_mean = vapply(cens_run, mean, numeric(1)),
  cens_sd   = vapply(cens_run, sd, numeric(1)))

fit_one_run <- function(d, formula, dist) {
  fit <- tryCatch(survreg(formula, data = d, dist = dist),
                  error = function(e) NULL,
                  warning = function(w) NULL)
  if (is.null(fit)) return(NULL)  # failed / no convergence
  se <- sqrt(diag(vcov(fit)))[names(coef(fit))]
  c(coef(fit), scale = fit$scale,
    setNames(se, paste0("se_", names(se))))
}
fit_model_across_runs <- function(df, formula, dist) {
  est <- lapply(split(df, df$run_id), fit_one_run,
                formula, dist)
  failed <- vapply(est, is.null, logical(1))
  out <- as.data.frame(do.call(rbind, est[!failed]))
  attr(out, "n_failed") <- sum(failed)
  out
}
f_full   <- Surv(time, status) ~ days + age + nr_comor + damage
f_no_age <- Surv(time, status) ~ days + nr_comor + damage
fit_all <- function(f, dist)
  lapply(simulated_data, fit_model_across_runs, f, dist)

model_estimates <- list(
  rq1_loglogistic = fit_all(f_full, "loglogistic"),
  rq2a_weibull    = fit_all(f_full, "weibull"),
  rq2b_no_age     = fit_all(f_no_age, "loglogistic"),
  censoring_summary = censoring_summary)
saveRDS(model_estimates, "model_estimates.rds")
```

Response to Feedback

Feedback received: "Report the mean value of the censoring ratio and the standard deviation across runs for each simulation scenario ($n = 200, n = 500$; 5%, 10%)."

How it was addressed: Censoring in our simulation comprises two distinct mechanisms: (i) random uninformative loss to follow-up prior to study conclusion ($L \sim \text{Unif}(0, c_{\max})$ with $L < \min(T, \tau)$), calibrated to 5% and 10%; and (ii) administrative censoring at the $\tau = 90$-day study horizon ($T \geq 90$). For each run across all $R = 2000$ simulation runs per scenario, we evaluated both the empirical loss to follow-up rate (status $= 0 \wedge$ time < 90) and the total composite censoring ratio (status $= 0$).

As shown in Table 3, empirical loss to follow-up closely matches our calibration targets ($5.00\% \pm 1.54\%$ and $10.04\% \pm 2.08\%$ for $n = 200$; $5.01\% \pm 0.96\%$ and $10.04\% \pm 1.39\%$ for $n = 500$). Administrative censoring accounts for the remaining censored cases ($\sim 37.8\%$ in the 5% scenario; $\sim 34.5\%$ in the 10% scenario), yielding overall composite censoring proportions of approximately 42.8% and 44.6%. The across-run standard deviations decrease by a factor of $\sqrt{500/200} \approx 1.58$ from $n = 200$ to $n = 500$, as expected from binomial sampling variability. Both summary metrics have been integrated into the simulation outputs in `hw2_model_fitting.R`.

Table 3: Empirical loss to follow-up and total censoring ratio across 2000 simulation runs (mean and std).

		Empirical Loss to Follow-up		Total Censoring Ratio	
Sample size (n)	Target Loss	Mean	Std. Dev.	Mean	Std. Dev.
200	5%	0.0500 (5.0%)	0.0154 (1.5%)	0.4286 (42.9%)	0.0351 (3.5%)
200	10%	0.1004 (10.0%)	0.0208 (2.1%)	0.4456 (44.6%)	0.0357 (3.6%)
500	5%	0.0501 (5.0%)	0.0096 (1.0%)	0.4273 (42.7%)	0.0220 (2.2%)
500	10%	0.1004 (10.0%)	0.0139 (1.4%)	0.4452 (44.5%)	0.0227 (2.3%)

Suggestions not adopted: None; the feedback was adopted in full.